SLOT 3 · FROM PRACTICE TO STANDARDS

The W3C Data Façades Community Group

What we are standardising, and what comes next: the research agenda and the engineering agenda.

How we got here

2021
Façade-X idea + SPARQL Anything
SEMANTiCS

2022
Materialisation approaches
Semantic Web J.

2023
The KGC method
ACM TOIT

2025
W3C Data Façades CG

2026
Theory of Façade-X data access
TGDK

Oct 2026
SPARQL Anything v1.3.0
Specs release 0.1

From one paper and one engine to **a community, several implementations, and a draft standard**.

Why a standard?

More than one engine

Façade-X is now implemented beyond SPARQL Anything. The same query should give the same answer everywhere.

Shared semantics

What does a CSV, a JSON number, an XML attribute become? Today the answer is "what the engine does".

Trust for adopters

A stable, versioned reference lowers the risk of building on the approach.

What the CG is writing

Model
containers, slots, values 0.1

Schema vocabulary
`fx:` terms used in the façade 0.1

Engine vocabulary
`fxe:` options, e.g. `fxe:csv.headers` 0.1

CSV
mapping 0.1

JSON
mapping 0.1

XML
mapping 0.1

Design
principles in progress

SPARQL data access
how a query reaches a façade 0.1

HTTP / protocol access
remote resources, APIs in progress

Primer
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Test suite
: input files and expected RDF, per format

How we work

In the open

Specs, issues and meeting minutes on GitHub. Anyone can read and comment.

Issue-driven

Each design decision is an issue: discussed in meetings, recorded in the spec.

Versioned releases

A latest release, a working draft, and frozen snapshots per version.

The research agenda

Theory beyond BGPs

Satisfiability is settled for basic graph patterns. Next: full SPARQL, and using it to prune and optimise.

Large data, small memory

Evaluate queries over big files as streams, without loading the whole façade.

Federation

Plan and optimise queries that join many façades and remote sources.

Types and entailment

Should `1775` match `"1775"^^xsd:decimal`? Term equality vs value-based matching.

Expressive power

Beyond context-free formats: what can, and should, a façade represent?

Façades and LLMs

One uniform shape for any source: a target for query generation and KG pipelines.

The engineering agenda: frameworks we need

Conformance

A shared test suite and a runner: any engine can check itself against the spec.

More engines

Façade-X inside more SPARQL engines and KG platforms, not only as a separate tool.

More formats

New mappings, shared across engines rather than re-invented in each.

Developer tooling

Editors, query debugging, inspecting a façade, validating it with SHACL.

Pipelines

Façade queries as reusable steps in ELT and data-space workflows.

A cookbook

Good practices and patterns, from the people who use it in production and teaching.

Get involved

Join the CG

Free, open to anyone with a W3C account.
w3.org/community/facade-x

Contribute

Comment on issues, propose a format mapping, add tests, report how you use it.
github.com/w3c-facade-x

Use and implement

Build on SPARQL Anything, or implement Façade-X in your engine.
sparql-anything.cc

Teach it

Reuse today's notebooks and slides in your courses.
w3c-facade-x.github.io/iswc2026-tutorial

Discussion

1. Which **formats** matter most to you, and which are missing?
2. What would stop you from **adopting** Façade-X in your work?
3. Would you **implement** it in your engine or platform? What would you need?
4. Which item of the **research agenda** should come first?

THANK YOU

Join us

W3C CG · w3.org/community/facade-x

Specs · github.com/w3c-facade-x

SPARQL Anything · sparql-anything.cc

Tutorial · w3c-facade-x.github.io/iswc2026-tutorial